

BRAC UNIVERSITY
Department of Computer Science and Engineering

CSE 260: Digital Logic Design
Md. Arifur Rahman [MAR]

Assessment: **Assignment 3**
Deadline: **11:59 PM, December 2, 2022**

Semester: **Fall 2022**
Full Marks: **10**

No late submission will be considered.

*This assignment must be hand-written. Show **all** steps in **all** questions.*

*Make sure that your circuit is **efficient**, that is, you should use the **lowest** number of components.*

You may use external gates if required.

Adder/Subtractor

1. **Design** a voter counting system using necessary full and parallel adder(s) for:
 - a. 13 people
 - b. 16 people

Lecture 7, Slides 27-29

2. **Design** a BCD to Excess-5 system using necessary parallel adder(s).

Lecture 7, Slide 31

3. Consider **A** to be a 4 bit number. **Design A-3** (3 subtracted from A) using a 4 bit parallel adder.

Lecture 7, Slide 24

Decoders

4. **Implement** the function $F(A,B,C,D) = \sum(0, 1, 4, 5, 7, 9, 10, 13)$ using:
 - a. 3:8 decoder(s) only
 - b. 2:4 decoder(s) only
 - c. both 3:8 and 2:4 decoders

Lecture 8, Slides 13-16

5. **Implement** the function $F(A,B,C,D,E) = \sum(0,1,2,7,8,10,11,13, 15,18,21,24,25)$ using:
 - a. 4:16 decoder(s) only
 - b. 2:4 decoder(s) only

Lecture 8, Slides 13-16

Multiplexers

6. **Construct** the following functions using a single 8:1 MUX:
 - a. $F(a,b,c) = \sum(0, 3, 5, 6, 7)$
 - b. $F(a,b,c,d) = \sum(0, 3, 5, 6, 7, 9, 11, 13, 15)$

a. Lecture 8, Slide 39

b. Lecture 8, Slides 40-45

7. **Implement** the function $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$ using:
 - a. a single 16:1 MUX
 - b. a single 8:1 MUX

a. Lecture 8, Slide 39

Resources:

0. Slides: [Class Folder](#)

1. Adder/Subtractor: refer to this [adder/sub playlist](#)

a. "Cascading Adders | Voting System Example" video contains several examples including 13 people voting system

b. also checkout [BCD-Excess 5](#) video

2. Decoder: refer to this [decoder playlist](#) or the [MSI Playlist](#)

3. MUX: [MSI playlist](#)